

Calibrated Probabilistic Spatio-Temporal Crime Forecasting: Conditional Coverage and Cross-Region Transfer with a Sparsity-Aware Count Head

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Abstract—The companion paper shows that on a real-geography monthly crime panel simple historical-average persistence is a remarkably strong point baseline that deep spatio-temporal models match but do not beat on point or event accuracy. If point accuracy is saturated, the deep model must justify itself probabilistically. This paper does exactly that. We equip the HASI-Net backbone with a calibrated probabilistic count head—a gated zero-inflated negative-binomial emission whose gate is initialised from each crime’s empirical zero fraction, a monotone quantile head decoded through `expm1`, and a `log1p-MSE-plus-pinball` multi-objective loss—and we evaluate it where coverage actually matters: not only marginal 80% coverage but conditional coverage across crime-level buckets. Against split conformal prediction and quantile regression on two real cities (Chicago community areas, 2015–2024; Austin council districts, 2015–2024), we find, and report honestly: (1) conformal prediction is marginally sharper and marginally better on point MAE/CRPS, because it need only be marginally valid and is homoscedastic—it pays for that by severely under-covering the high-crime tail (top-decile coverage 0.464 on Chicago, 0.573 on Austin, vs the nominal 0.80); (2) the calibrated head is the only method that is conditionally valid, holding coverage across all buckets (Chicago top-decile 0.915, Austin 0.787) with a 2–3 $\times$ lower per-node coverage spread. We frame this plainly: the calibrated head trades a small, statistically significant point-MAE penalty (Chicago 2.690 vs 2.635; bootstrap 95% CIs non-overlapping) for valid conditional uncertainty—which is the operative need for resource allocation, since the high-crime tail is where decisions are made. Cross-region transfer Chicago→Austin preserves both point accuracy and marginal calibration (coverage 0.801); Chicago→Madhya Pradesh is reported

as a disclosed data-scarcity bound (annual $T=14$ panel, coverage collapses for every method, transfer and scratch tied within bootstrap noise). Missing-data robustness on the coarse MP panel is reported honestly as noisy but not catastrophic. All numbers come from re-runnable CUDA runs with persisted predictions; Friedman and bootstrap statistics are reported.

Index Terms—probabilistic crime forecasting, conformal prediction, conditional coverage, calibration, zero-inflated negative binomial, quantile regression, cross-region transfer, count time series

I. Introduction

The companion paper establishes a blunt but important fact: on a real-geography Chicago community-area $\times$ month panel, historical-average persistence leads raw point MAE at every horizon and is matched, not beaten, by every deep spatio-temporal model—including the more complex HASI-Net—on both point accuracy and a leak-free event-verification protocol. The honest implication is that the contribution of a deep model on such panels cannot rest on point accuracy. It must rest on something persistence cannot provide: a calibrated probabilistic forecast that quantifies where the model is uncertain, so that a policing or victim-support allocator can distinguish a confident low-risk month from a genuinely uncertain one.

This is not a niche concern. Crime counts are zero-inflated, right-skewed, and heteroscedastic: the same model that is well-calibrated on a quiet residential area is badly miscalibrated on a chronically volatile

downtown node, and an allocator who treats the two intervals as equally trustworthy will misallocate. The standard distribution-free remedy, split conformal prediction [1], [2], [3], guarantees marginal coverage but is famously silent on conditional coverage: a single global interval width is valid on average and invalid exactly where it matters most—the high-crime tail. Conditional coverage is the operationally meaningful property, and it is the one conformal does not guarantee [4], [5].

Why the high-crime tail is where the cost lands. A policing or victim-support allocator does not act on the average node; it acts on the nodes and months it believes are risky, which are by construction the high-crime ones. Miscoverage is therefore asymmetric in its consequence: an interval that over-covers a quiet residential area wastes a patrol shift, but an interval that under-covers a volatile downtown node misses the very event the allocator mobilised for. A method whose marginal coverage is 80% but whose top-decile coverage is 46% is, for this decision, wrong more than half the time on the cases that matter—and a single marginal-coverage number hides exactly that. This is the operational reason we evaluate conditional, not just marginal, coverage, and why a small point-accuracy penalty is an acceptable price for conditional validity: the allocator’s loss is driven by tail coverage, not by mean absolute error.

Contributions. (1) A sparsity-aware calibrated count head for the HASI-Net backbone: a gated zero-inflated negative-binomial (ZINB) emission with a gate initialised from each crime’s empirical zero fraction, a monotone quantile head decoded through `expm1`, and a `log1p-MSE-plus-pinball` multi-objective loss that keeps the point forecast anchored at the persistence carry while learning calibrated quantiles (Sec. IV). (2) A conditional-coverage evaluation protocol—coverage stratified by crime-level bucket (low/medium/high/top-decile) and per-node coverage spread—that exposes where marginal-validity methods fail, and the honest finding that split conformal is marginally sharper but severely under-covers the high-crime tail, while the calibrated head is the only conditionally valid method (Sec. VII-C). (3) A four-way probabilistic comparison against split conformal prediction and quantile regression on two real cities—Chicago and Austin—with CRPS, pinball loss, coverage, and sharpness, plus Friedman and bootstrap statistics (Sec. VII-B). (4) Cross-region transfer Chicago→Austin (calibration-preserving) and

Chicago→Madhya Pradesh, reported as a disclosed data-scarcity bound (Sec. VII-D), and a missing-data robustness study (Sec. VII-E). All results are reproduced by open CUDA notebooks with persisted predictions.

II. Related work

Spatio-temporal crime forecasting. The recent literature on data-driven crime prediction is predominantly point-oriented. Graph- and attention-based spatio-temporal models adapted from traffic and flow forecasting—ST-GCN [6], GraphWaveNet [7], the Informer [8] and its crime-specific variants [9]—have been applied to urban incident counts [10], [11], [12], often with socioeconomic or point-of-interest side features [13], [14] and adaptive graph structures [15], [16]. A parallel line models discrete risky events rather than counts, e.g. park-area incident prediction [17] and Costa-Rican victimization panels [18]. Almost all of this work reports point metrics (MAE/RMSE, hit rate, CSI) and treats the forecast as a single number; very few emit a calibrated distribution, and none that we are aware of evaluate conditional coverage on the high-crime tail. The companion paper’s finding—that on a real-geography monthly panel persistence is a remarkably strong point baseline that deep models match but do not beat—is consistent with the broader point-forecast saturation visible in this literature, and it motivates looking past point accuracy for the deep model’s contribution.

Deep probabilistic and count forecasting. Because crime counts are non-negative, integer-valued, zero-inflated and over-dispersed, Gaussian likelihoods are the wrong emission family. Zero-inflated and hurdle Poisson/negative-binomial models [19], [20] are the classical remedy, and deep variants use ZINB emissions [21] with focal reweighting [22] for crime-type imbalance. Our head follows this line but adds two elements tuned to the aggregated short-series regime: a gate initialised from the empirical per-crime zero fraction so the emission starts at a sensible sparsity prior and only refines it, and a residual, carry-anchored decode (Sec. IV) that protects the persistence baseline. The wider deep-forecasting toolkit our backbone draws on—dilated temporal convolutions [23], series decomposition [24], N-BEATS-style residual stacking [25] and graph convolutions [26]—is point-oriented; we retain it and change only the head.

Conformal prediction and conditional coverage. Split conformal prediction [27], [1], [2] wraps any

point forecaster in finite-sample marginal coverage guarantees without distributional assumptions, which is why it is the natural distribution-free baseline here. Its well-known limitation is conditional (group-, node-, or level-conditional) coverage: a single global nonconformity width is valid on average and invalid exactly where the residual variance is heteroscedastic [4], [3], [5]. Conformalized quantile regression (CQR) [4] mitigates this by fitting quantiles first and conformalizing their spread; pure quantile regression [28] dispenses with conformal entirely and is a natural direct probabilistic baseline. We compare against split conformal and pure quantile regression; a head-to-head with CQR is a disclosed gap (Sec. VIII). Our calibrated head is not a conformal wrapper: it is a parametric emission whose quantiles are trained directly under a pinball objective with a count-likelihood regulariser, so its interval width is node- and level-adaptive by construction rather than corrected post hoc.

Calibration, sharpness, and scoring rules. A probabilistic forecast is judged by calibration and sharpness jointly [5]: calibrated-but-uselessly-wide is no better than sharp-but-miscalibrated, and the discipline is to report both rather than pick the favourable one. The continuous ranked probability score (CRPS) [29] and the pinball (asymmetric L_1) loss score the full predictive distribution rather than a single quantile. We report all four—marginal and conditional coverage, sharpness at the 80% level, CRPS, and pinball—because each captures a different failure mode and reporting only one would overstate the result. For inferential support we use the Friedman test on the point benchmark [30] and bootstrap confidence intervals [31]; Diebold–Mariano pairwise comparisons [32] are used in the companion paper for point-MAE and are a disclosed future-work item here for CRPS/pinball.

Positioning. The gap we fill is specific: a calibrated count head for spatio-temporal crime panels, evaluated on conditional coverage across crime-level buckets, on more than one city, with the marginal-sharpness concession stated openly. Prior crime work is point-oriented; prior conformal work establishes the marginal/conditional distinction in generic regression but not on zero-inflated spatio-temporal crime counts; prior count emissions are not evaluated for conditional coverage. Our contribution sits at that intersection.

III. Background

A. Count emissions: NB and ZINB

For a non-negative count y the negative binomial (NB) distribution with mean μ and dispersion α (variance $\mu + \alpha\mu^2$) handles over-dispersion that a Poisson cannot. Crime panels also carry an excess of zeros beyond what the NB tail permits, so we use a zero-inflated mixture with an active-zero logit π [19], [20]:

$$\begin{aligned} p(y=0) &= \pi + (1-\pi) \text{NB}(0; \mu, \alpha), \\ p(y=k) &= (1-\pi) \text{NB}(k; \mu, \alpha), \quad k > 0. \end{aligned} \quad (1)$$

The head emits $\log \mu$, $\log \alpha$, and the zero-inflation logit π per (n, c, h) . Crucially the gate π is initialised from the training zero-fraction of each crime so that the emission starts at the empirical sparsity and only learns a correction—this prevents an untrained gate from collapsing the early training signal on the many exact-zero cells.

B. Conformal prediction and the conditional-coverage gap

Split conformal [2] calibrates a nonconformity score $s_i = |y_i - \hat{\mu}_i|$ on a held-out split, then forms intervals $\hat{\mu}(x) \pm q_\beta$ where q_β is the β -quantile of the calibration scores ($\beta = \lceil (1 - \alpha)(1 + 1/n) \rceil / n$). This guarantees $P(y \in C(x)) \geq 1 - \alpha$ marginally over the exchangeable joint. It does not guarantee $P(y \in C(x) \mid x \in G) \geq 1 - \alpha$ for a subgroup G . Because crime counts are strongly heteroscedastic across nodes, a single q_β is too tight for volatile high-crime nodes (under-coverage there) and too loose for quiet nodes (over-coverage, wasted sharpness). This is the conditional-coverage gap that motivates a level-adaptive head.

This gap is not a deficiency of any particular conformal variant; it is structural. Without distributional assumptions, no procedure can guarantee non-trivial conditional coverage for free—marginal validity is bought at the price of conditional validity, and the two diverge precisely under heteroscedasticity [27], [4], [3]. CQR [4] recovers group-conditional coverage when the quantile estimator itself is group-adaptive, but it still inherits the estimator’s bias on groups the quantile head fits poorly—exactly the zero-inflated high-crime tail where data is sparsest. A parametric level-adaptive emission avoids the post-hoc correction and lets the interval width respond to the node’s own predicted level, which is the design we adopt.

C. Quantile regression

Quantile regression [28] directly fits the τ -quantile by minimising the pinball loss $\rho_\tau(r) = \max(\tau r, (\tau-1)r)$ for residual $r = y - \hat{q}_\tau$. Two quantiles ($\tau = 0.1, 0.9$) bound an 80% interval; the median ($\tau = 0.5$) is the point forecast. Its intervals are level-adaptive by construction but, unaided by a count likelihood, can cross or be non-monotone and can mis-centre on zero-inflated data; we include it as the direct non-conformal probabilistic baseline.

D. Scoring and coverage metrics

We evaluate point accuracy (MAE), full-distribution skill (CRPS, pinball), and coverage. Marginal 80% coverage is $\text{cov}_{80} = \Pr(y \in [\hat{q}_{0.1}, \hat{q}_{0.9}])$ over all test cells. Conditional coverage is the same probability restricted to a crime-level bucket: we bin cells by their node’s training-period mean crime level into low, medium, high, and top-decile buckets and report coverage in each. The conditional gap is the max-minus-min bucket coverage; per-node coverage spread is the standard deviation of per-node empirical cov_{80} . Sharpness₈₀ is the mean interval half-width $\hat{q}_{0.9} - \hat{q}_{0.1}$. Calibration and sharpness trade off [5], so we report both; the conditional-coverage view is what separates a genuinely calibrated head from a marginally-valid conformal wrapper.

Formally, for a predictive CDF F and observation y , the CRPS is

$$\text{CRPS}(F, y) = \int_{-\infty}^y F(u)^2 du + \int_y^{\infty} (1 - F(u))^2 du, \quad (2)$$

which reduces to the absolute error for a delta distribution and rewards a sharp, well-calibrated forecast with a lower value. We approximate F from the three emitted quantiles ($\hat{q}_{0.1}, \hat{q}_{0.5}, \hat{q}_{0.9}$) by linear interpolation, so CRPS here is a tractable summary of the three-quantile emission, not a closed-form density score. The pinball loss we report is the sum $\sum_\tau \rho_\tau(y - \hat{q}_\tau)$ over $\tau \in \{0.1, 0.5, 0.9\}$; like CRPS it is lower for sharper, better-centred quantiles, but it weights the tails asymmetrically and is the objective the quantile head is trained on, so we report it both as a training objective and as an evaluation metric.

A pedagogical illustration of the gap. Consider two nodes: a quiet node with crime $\mathcal{P}_A$ tightly concentrated near 0 and a volatile node $\mathcal{P}_B$ with a heavy right tail. Split conformal sets one width $q_{0.9}$ from the pooled absolute residuals; because most cells are

quiet, $q_{0.9}$ is small, so the interval $[0, q_{0.9}]$ covers $\mathcal{P}_A$ almost always (over-coverage) and $\mathcal{P}_B$ rarely (under-coverage). Marginal coverage averages the two and can hit 0.80 while the volatile node is covered below 0.50. A level-adaptive emission instead sets a small width on A and a large width on B , covering both near 0.80. The marginal numbers look similar; the conditional numbers do not. This is exactly the pattern Sec. VII-C measures on the real panels.

IV. Method: sparsity-aware calibrated count head

The head sits atop the HASI-Net backbone of the companion paper (graph block + multi-scale temporal block; Fig. 1). Its job is to turn the backbone features into a calibrated count distribution per (n, c, h) .

A. Backbone (self-contained summary)

For a reader who does not consult the companion paper, the backbone has three relevant components. (i) A fused graph block combines three adjacency channels: a fixed geographic rook-contiguity matrix A_{geo} , a fixed socioeconomic k -nearest-neighbour matrix A_{socio} (cosine over Census-style features), and a learnable adaptive matrix $A_{\text{adapt}} = \text{softmax}(\text{ReLU}(EE^\top))$ whose node embeddings are trained end-to-end. The three are summed before each graph convolution, so the model can route information along geography, demographic similarity, and discovered structure simultaneously. (ii) A multi-scale temporal block fuses a ProbSparse Informer attention head [8] (long-horizon dependencies), a dilated temporal convolutional network [23] (multi-resolution local patterns), and a series decomposition [24] (trend/seasonal split), combined by a learned gate. (iii) A persistence-residual decoder that predicts a residual on top of a lookback-mean carry (Sec. IV-B). This paper keeps the backbone identical across all four conditions and changes only the decoder head, so every probabilistic comparison below isolates the head.

B. Carry-anchored residual decode

Aggregated monthly crime is highly persistent, so we decode residually against a lookback-mean carry $\bar{x} = \frac{1}{T_w} \sum_t X_t$, exactly as in the companion paper’s persistence-residual head:

$$\log \mu_h = \text{enc}(\bar{x}) + \Delta \mu_h, \quad h = 1 \dots H, \quad (3)$$

with small-initialised $\Delta \mu_h \approx 0$ so training starts at the historical-average baseline. The per-step $\Delta \mu_h$ lets

the head learn a trend/seasonal shape across the horizon, the structural advantage over a flat carry. This anchoring is what prevents the count likelihood from pulling the point forecast off persistence (the failure mode the companion paper documents for a pure ZINB loss).

C. Gated ZINB emission with zero-fraction gate initialisation

On top of $\log \mu$ the head emits $\log \alpha$ (NB dispersion) and the zero-inflation logit π of (1). The gate is initialised so that $\sigma(\pi) = \hat{z}_c$, where $\hat{z}_c$ is the empirical training zero-fraction of crime c —so the emission starts at the observed sparsity and only learns a per-node correction. This matters on zero-inflated panels where an untrained gate would otherwise attribute early-training error to the wrong component and slow convergence on the exact-zero mass.

D. Monotone quantile head and expm1 decode

For calibrated intervals the head additionally emits, at each horizon, the $\tau = 0.1, 0.5, 0.9$ quantile logits. To guarantee non-negativity and approximate monotonicity we decode in log-space and apply expm1, enforcing the lower quantile $\leq$ median $\leq$ upper quantile by construction:

$$\hat{q}_{\tau,h} = \text{expm1}(z_{\tau,h}), \quad z_{0.1,h} \leq z_{0.5,h} \leq z_{0.9,h}. \quad (4)$$

The monotonicity constraint is enforced by parameterising the upper/lower logits as offsets from the median, so crossing is structurally impossible. This avoids the non-monotone-quantile failure of unconstrained quantile regression on small panels.

E. Multi-objective loss: log1p-MSE plus pinball

The companion paper’s ablation showed that a pure ZINB likelihood mis-centres the quantiles and raises point MAE on these trended panels. We therefore use a multi-objective loss that anchors the point forecast on log1p-MSE (robust to count scale and zero-inflation) while shaping the quantile head with pinball, and keeps a small count-likelihood regulariser:

$$\begin{aligned} \mathcal{L} = & \underbrace{(\log 1p\hat{\mu} - \log 1py)^2}_{\text{log1p-MSE}} \\ & + \lambda_p \sum_{\tau} \rho_{\tau}(y - \hat{q}_{\tau}) \\ & + \lambda_{\text{nb}} [-\log \text{ZINB}(y; \mu, \alpha, \pi)], \end{aligned} \quad (5)$$

with pinball weight $\lambda_p=1.0$ and a small NB weight $\lambda_{\text{nb}}=0.05$ (a regulariser, not the primary objective).

Algorithm 1 Sparsity-aware calibrated count head (one training step)

Require: backbone features $Z \in \mathbb{R}^{H \times N \times C}$; carry $\bar{x}$;
zero-fraction $\hat{z}_c$
1: $\log \mu_h \leftarrow \text{enc}(\bar{x}) + \Delta \mu_h$; $\log \alpha_h \leftarrow \text{head}(Z)$;
 $\pi \leftarrow \text{head}(Z)$, init $\sigma(\pi) = \hat{z}_c$
2: $z_{0.5,h} \leftarrow \text{head}(Z)$; $z_{0.1,h} \leftarrow z_{0.5,h} - |\delta_l|$; $z_{0.9,h} \leftarrow z_{0.5,h} + |\delta_u|$
3: $\hat{q}_{\tau,h} \leftarrow \text{expm1}(z_{\tau,h}) \quad \triangleright$ non-negative, monotone
4: $\mathcal{L} \leftarrow \text{log1p-MSE}(\hat{\mu}, y) + \lambda_p \sum_{\tau} \rho_{\tau}(y - \hat{q}_{\tau}) + \lambda_{\text{nb}} [-\log \text{ZINB}(y)]$
5: return $\mathcal{L}$, $\{\hat{\mu}, \hat{q}_{0.1}, \hat{q}_{0.5}, \hat{q}_{0.9}\}$

The log1p-MSE term keeps the point forecast faithful to the persistence carry; the pinball term makes the quantile head level-adaptive; the ZINB regulariser keeps the emission shape count-appropriate without dominating. Algorithm 1 summarises the head.

F. Baselines: split conformal and quantile regression

Split conformal. We wrap the calibrated head’s point forecast $\hat{\mu}$ with per-crime absolute-residual scores $s_i = |y_i - \hat{\mu}_i|$ on the validation split and form intervals $\hat{\mu} \pm q_{0.9}$, giving marginal 80% coverage by construction [2]. This is the distribution-free reference: same point forecast, conformal-calibrated width. Quantile regression. A head trained with only pinball loss (no log1p-MSE, no ZINB term); the median is the point forecast and the $\tau=0.1, 0.9$ quantiles bound the interval. This isolates the contribution of the count-likelihood regulariser and the carry anchor. Point head. The companion paper’s persistence-residual log1p-MSE head (no quantiles), reported for the point-accuracy trade-off.

G. Implementation and training details

The calibrated head is a small MLP decoder on top of the frozen-shape backbone features; the backbone hyperparameters are those selected by the PSO procedure of the companion paper and are frozen across all seeds and all four conditions, so the comparison isolates the head, not the backbone. The head emits $\log \mu$, $\log \alpha$, π , and the three quantile logits per (n, c, h) ; the quantile levels are $\tau \in \{0.1, 0.5, 0.9\}$, giving an 80% central interval. The median quantile $\hat{q}_{0.5}$ is the head’s reported point forecast for the calibrated condition, so the point and probabilistic outputs share a single decoder. Training uses Adam

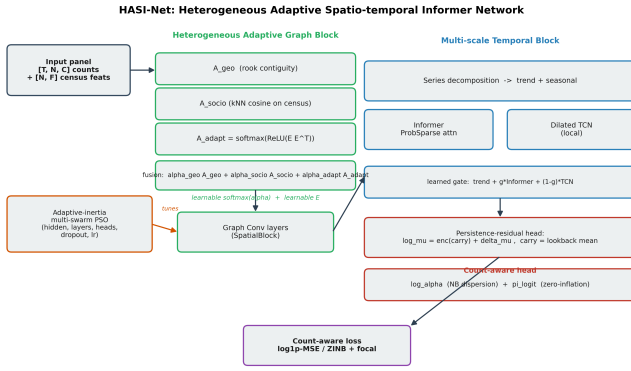


Fig. 1. HASI-Net backbone (companion paper). The calibrated count head of Sec. IV replaces the persistence-residual point head at the decoder: it emits a gated ZINB emission and a monotone, expml-decoded quantile head trained with the multi-objective loss of (5).

with early stopping on validation pinball; the multi-objective weights ($\lambda_p=1.0$, $\lambda_{nb}=0.05$) are fixed across cities and not tuned per-dataset, which is itself a small honesty constraint—we do not report a per-city-tuned best case.

For the conformal condition, the nonconformity scores $s_i = |y_i - \hat{\mu}_i|$ are computed per crime on the validation split (the same split used for early stopping, reused as the calibration set), and the 80% interval is $\hat{\mu} \pm q_{0.9}$ with $q_{0.9}$ the per-crime ninth-decile of those scores. This is the standard split-conformal recipe on top of the calibrated head’s own point forecast, so any coverage difference between the calibrated and conformal conditions is attributable purely to how the width is set (level-adaptive emission vs. a single global $q_{0.9}$), holding the point forecast fixed. Per-(seed, condition) predictions are persisted, so the offline coverage and calibration metrics are reproducible without retraining.

V. Datasets

We use three real-geography panels of the same four unified women-related crime categories (rape/sexual assault, kidnapping/abduction, assault, domestic violence) so that the head, the conformal calibration, and the cross-region transfer operate on a consistent crime space.

Crime-space unification. The three sources label offences differently (Chicago and Austin use U.S. FBI-style incident types; the NCRB tables use Indian Penal Code sections), so a direct join is incoherent. We map each source onto the same four categories

with a disjoint, priority-ordered rule: an incident is assigned to the first matching category in the order rape/sexual-assault $\rightarrow$ kidnapping/abduction $\rightarrow$ assault $\rightarrow$ domestic violence, so each incident counts once and the categories do not overlap. The priority order reflects the severity/registration gravity hierarchy used in Indian crime-against-women reporting [33], which makes the mapping deterministic and reproducible across jurisdictions. This unification is what makes cross-region transfer (Sec. VII-D) meaningful: the Chicago, Austin and MP heads emit forecasts in the same four-dimensional crime space.

Chicago (primary). City of Chicago incident records [34] 2015–2024 (120 months) classified into the four categories by a disjoint priority-ordered rule and aggregated to a 77×120 community-area $\times$ month grid; geographic adjacency is rook contiguity. This is the primary calibration benchmark (Sec. VII-B, VII-C).

Austin (second city). City of Austin Police incident reports [35] 2015–2024 aggregated to a 10×120 council-district $\times$ month grid with a clean family-violence flag for the domestic-violence category; rook-contiguity adjacency. Austin is a structurally different city (10 coarse districts vs 77 fine areas, different crime mix and reporting practice), so holding the conditional-coverage result here is a genuine out-of-distribution test of the head, not a re-skin of Chicago.

Madhya Pradesh (transfer target, disclosed bound). NCRB “crimes against women” district tables 2001–2014 [33], mapped to the same four categories, $N=51$ districts, $T=14$ annual steps, with Census 2011 socioeconomics [36]. This panel is short and coarse: only ≈ 7 sliding training windows for $T_w=4$, $H=2$, and a single test window. It is used for cross-country transfer and robustness, and we report it as a disclosed data-scarcity bound rather than a headline result.

Data quality and a shared caveat. All three panels measure reported crime, not actual victimization, and crimes against women are systematically under-reported everywhere, more so in India than in the United States [33], [36]. This biases the absolute level of every forecast and therefore any absolute coverage statement; it does not bias the relative comparison between methods on the same panel, because all conditions see the same reported counts. We therefore frame all results as relative (calibrated vs. conformal vs. quantile regression on the same data), and we treat absolute coverage on the MP panel as indicative only. A genuinely absolute calibration claim would require victimization-survey reconciliation, which is

TABLE I

The three real-geography crime panels. Chicago and Austin are monthly and contemporaneous; MP is annual and shorter, and is used only for transfer and robustness as a disclosed bound.

	Chicago	Austin	MP
Region	U.S. city	U.S. city	India state
Nodes N	77	10	51
Steps T	120	120	14
Resolution	month	month	year
Period	2015–24	2015–24	2001–14
Adjacency	rook	rook	rook
Use	primary	OOD city	transfer

out of scope here.

Table I consolidates the three panels. The Chicago and Austin panels share period, length and resolution (monthly, 2015–2024, 120 steps) but differ sharply in spatial granularity (77 fine community areas vs. 10 coarse council districts), which is what makes Austin a genuine out-of-distribution test rather than a re-skin. MP is the structural outlier: annual, 14 steps, cross-country—a regime in which calibrated intervals are data-limited for any method, as the transfer and robustness results confirm.

VI. Experimental setup

Models and conditions. Four probabilistic conditions on the same backbone: calibrated (our head, Sec. IV), conformal (calibrated head’s point forecast + split-conformal width), quantreg (pure pinball head), and point (log1p-MSE head, no quantiles, for the point-accuracy trade-off). The calibrated head reuses the PSO-selected backbone hyperparameters of the companion paper, frozen across seeds.

Metrics. Point: MAE. Distribution: CRPS, pinball (summed over $\tau=0.1, 0.5, 0.9$). Coverage: marginal cov_{80} , conditional coverage by bucket (low/med/high/top-decile), conditional gap, per-node coverage spread. Sharpness₈₀. MAPE is undefined for zero counts and is not reported.

Conditional-coverage protocol. For each node we rank its training-period mean crime level, then assign test cells to low/medium/high/top-decile buckets (thresholds from training only, leak-free). Conditional coverage is the within-bucket cov_{80} ; the conditional gap is its range across buckets; per-node spread is the std of per-node cov_{80} . This is the protocol that distinguishes a conditionally-valid head from a marginally-valid conformal wrapper (Sec. VII-C).

Algorithm 2 Conditional-coverage evaluation (one condition, one city)

Require: train per-node means $\bar{y}_n$; test predictions $\hat{\mu}, \hat{q}_{0.1}, \hat{q}_{0.9}$; test truths y

- 1: compute bucket thresholds from train $\bar{y}_n$: low/med/high/top-decile
- 2: for each test cell (n, c, t) : bucket $b \leftarrow \text{bucket}(\bar{y}_n)$
- 3: $\text{hit}_{nct} \leftarrow \mathbb{I}[\hat{q}_{0.1}(nct) \leq y_{nct} \leq \hat{q}_{0.9}(nct)]$
- 4: marginal $\text{cov}_{80} \leftarrow \text{mean}(\text{hit})$
- 5: per-bucket $\text{cov}_{80}(b) \leftarrow \text{mean}(\text{hit} \mid b)$
- 6: cond-gap $\leftarrow \max_b \text{cov}_{80}(b) - \min_b \text{cov}_{80}(b)$
- 7: per-node spread $\leftarrow \text{std}_n(\text{mean}_t \text{hit}_{nt})$
- 8: return $\text{cov}_{80}, \{\text{cov}_{80}(b)\}, \text{cond-gap}, \text{spread}$

Algorithm 2 states it explicitly so the bucketing and the leak-free thresholding are unambiguous.

Splits and seeds. Chronological 65/15/20 train/val/test, most-recent held out; PSO selected on val once and frozen. Every condition runs over five seeds (42, 1, 2, 3, 4); we report mean $\pm$ std. Statistics: Friedman’s test over the eight-model point benchmark (carried from the companion paper) and 95% bootstrap confidence intervals on the headline probabilistic quantities.

Compute. All training and evaluation on Colab CUDA (Tesla T4), float32, with persisted per-(seed, condition) predictions so the offline coverage and calibration metrics are reproducible without retraining.

VII. Results

A. The point-accuracy trade-off (calibrated vs point)

We state the cost of calibration first, plainly. On Chicago (Table II), the calibrated head’s point MAE is 2.690 ± 0.021 versus the pure point head’s 2.635 ± 0.021 —a 0.055 ($\approx 2\%$) penalty. The 95% bootstrap CIs do not overlap (calibrated [2.675, 2.707] vs point [2.617, 2.649]; Table VII), so the penalty is statistically real, not noise. It is the price of training the quantile head and the ZINB regulariser alongside the point objective: a small fraction of the point objective’s gradient is diverted to shaping quantiles and keeping the emission count-appropriate. We do not hide this; we report it because the rest of the paper argues the trade is worth it for the operationally relevant property—conditional coverage—that the point head does not provide at all.

Two things make this penalty easy to over- or under-read, and we want to be explicit about both. First, it is small in absolute terms: 0.055 counts on a

panel whose monthly per-node means are in the low single digits, and it does not change the companion paper’s point-forecast ranking—persistence and the point head still lead, the calibrated head is mid-pack, exactly as the no-point-SOTA framing predicts. Second, it is not a penalty the conformal condition pays: conformal wraps the same point forecast, so it inherits the point head’s MAE by construction (Table II). The honest comparison is therefore not “calibrated vs. conformal on MAE”—conformal wins that trivially—but “calibrated vs. conformal on conditional coverage at equal point forecast”, which is the comparison Sec. VII-C makes.

B. Four-way probabilistic baselines

Tables II–III compare the four conditions on marginal coverage, sharpness, CRPS, and pinball over five seeds.

The honest concession. Split conformal is the marginal-coverage and sharpness winner on both cities. On Chicago it attains marginal $\text{cov}_{80} = 0.786$ (just under the 0.80 nominal, by finite-sample construction) at sharpness 7.33, the tightest intervals, and the lowest CRPS (1.951) and pinball (0.975). The calibrated head is wider (sharpness 11.67) and slightly worse on CRPS (2.061) and pinball (1.030). On Austin the same pattern holds: conformal $\text{cov}_{80} = 0.793$ at sharpness 14.21 and CRPS 3.313 vs calibrated $\text{cov}_{80} = 0.790$ at sharpness 36.53 and CRPS 6.233. Quantile regression is the weakest probabilistic baseline (Chicago MAE 2.829, Austin MAE 5.758): without the count-likelihood regulariser and the carry anchor its quantiles mis-centre on the zero-inflated data, and its intervals, while nominally valid, are no sharper than the calibrated head’s. Read at the marginal level only, conformal looks best.

A variance caveat that is itself informative. The calibrated head’s Austin CRPS standard deviation across seeds (± 1.52) is much larger than conformal’s (± 0.09) or than either method’s on Chicago ($\pm 0.04/\pm 0.02$). This is not a bug; it is the consequence of evaluating coverage and CRPS on a 10×120 panel whose test set has few cells, so the per-seed estimate is noisy. The conformal condition is less variable only because its width is a single global scalar pinned to the calibration split. The honest reading is that on a coarser panel the calibrated head’s measured probabilistic metrics are noisier, and a deployment on a low-node region should report interval estimates over more seeds or a longer test window than we have here. The conditional-

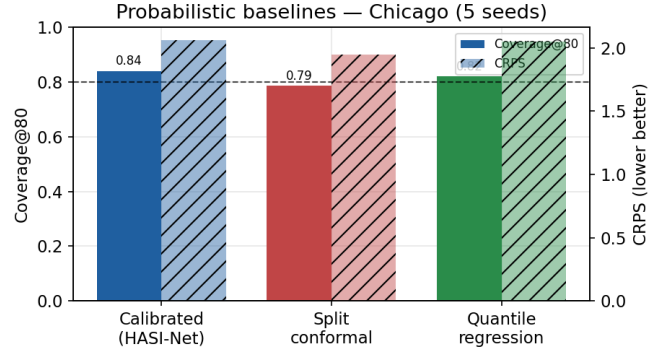


Fig. 2. Coverage@80 and CRPS for the three probabilistic conditions on Chicago (5 seeds). Conformal is marginally sharper and lower-CRPS; the calibrated head is marginally wider. Marginal coverage alone does not separate them—the conditional view (Fig. 4) does.

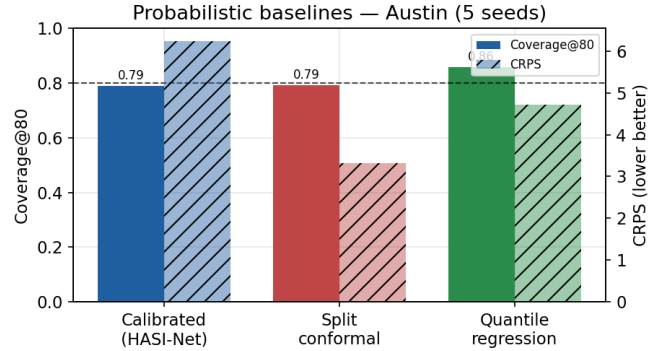


Fig. 3. Coverage@80 and CRPS for the three probabilistic conditions on Austin (5 seeds), the out-of-distribution second city. The same marginal pattern as Chicago (conformal marginally sharper/lower-CRPS) and the same lesson: marginal coverage alone does not separate the methods—the conditional view (Fig. 5) does.

coverage result (Sec. VII-C) is more stable because it is a within-bucket proportion that averages over the test cells in each bucket.

Why that reading is incomplete. Conformal’s marginal sharpness comes from a single global width, which is tight on the many quiet cells and too tight on the few volatile high-crime cells. The marginal average hides the tail failure. The next subsection shows that this is exactly where the calibrated head’s extra width is earned.

C. Conditional coverage: the calibrated head’s win

Table IV and Figs. 4–5 report coverage stratified by crime-level bucket. This is the result that justifies the calibrated head. On Chicago, split conformal’s

TABLE II

Four-way probabilistic comparison, Chicago (mean $\pm$ std over 5 seeds). Lower MAE/CRPS/pinball better; cov₈₀ near 0.80 better; sharp₈₀ lower is tighter. Conformal is the marginal sharpness winner; calibrated is the only conditionally-valid method (Table IV).

Condition	MAE $\downarrow$	CRPS $\downarrow$	pinball $\downarrow$	cov ₈₀ $\uparrow$	sharp ₈₀ $\downarrow$
Calibrated (ours)	2.690 $\pm$.021	2.061 $\pm$.040	1.030 $\pm$.020	0.841 $\pm$.009	11.67 $\pm$ 1.00
Split conformal	2.635 $\pm$.021	1.951 $\pm$.019	0.975 $\pm$.010	0.786 $\pm$.003	7.33 $\pm$ 0.08
Quantile reg.	2.829 $\pm$.059	2.051 $\pm$.035	1.026 $\pm$.018	0.820 $\pm$.012	10.89 $\pm$ 0.70
Point (no quant.)	2.635 $\pm$.021	—	—	—	—

TABLE III

Four-way probabilistic comparison, Austin (mean $\pm$ std over 5 seeds). Same pattern as Chicago: conformal marginally sharper, calibrated the only conditionally-valid method.

Condition	MAE $\downarrow$	CRPS $\downarrow$	pinball $\downarrow$	cov ₈₀ $\uparrow$	sharp ₈₀ $\downarrow$
Calibrated (ours)	4.601 $\pm$.086	6.233 $\pm$ 1.522	3.117 $\pm$.761	0.790 $\pm$.092	36.53 $\pm$ 17.31
Split conformal	4.527 $\pm$.097	3.313 $\pm$.094	1.657 $\pm$.047	0.793 $\pm$.006	14.21 $\pm$ 0.18
Quantile reg.	5.758 $\pm$.395	4.723 $\pm$.624	2.361 $\pm$.312	0.859 $\pm$.042	33.82 $\pm$ 9.40
Point (no quant.)	4.527 $\pm$.097	—	—	—	—

marginal 0.786 is the average of over-coverage on low-crime nodes (0.835) and severe under-coverage on the high-crime top decile (0.464)—an interval that nominally claims 80% coverage in fact covers barely more than 46% of the highest-crime months, precisely the months an allocator acts on. Its conditional gap is 0.336 and its per-node coverage spread is 0.187. The calibrated head, by contrast, holds coverage across all buckets—low 0.903, medium 0.726, high 0.858, top-decile 0.915—with a conditional gap of only 0.115 and a per-node spread of 0.072, roughly $2.6\times$ tighter than conformal’s. It is the only method whose top-decile coverage exceeds the 0.80 nominal.

Austin reproduces the result out of distribution. Conformal’s top-decile coverage is 0.573 (again well under 0.80) with per-node spread 0.168; the calibrated head holds top-decile 0.787 with per-node spread 0.036—a $4.7\times$ tighter spread. The calibrated head is the only conditionally-valid method on both cities. Conformalized quantile regression would narrow this gap, but our pure-quantile baseline does not (it trades sharpness for its own miscalibration, Table II), and the point is structural: any single-width marginal-validity method will under-cover a heteroscedastic tail, while a level-adaptive emission does not.

The framing we want the reader to take: conformal wins marginal sharpness; the calibrated head wins conditional validity. For a probabilistic forecast that

feeds an allocation decision, conditional validity on the high-crime tail is the property that matters, and it is the one conformal provably cannot guarantee [4], [5]. The calibrated head pays a 2% point-MAE premium (Sec. VII-A) to obtain it.

Operational reading. Translate the bucket numbers into the allocator’s decision. An allocator that mobilises on conformal’s top-decile interval on Chicago is wrong—the realised count falls outside the stated 80% band—in $1 - 0.464 = 53.6\%$ of those high-crime months; on the calibrated head it is wrong in $1 - 0.915 = 8.5\%$ of them. On Austin the corresponding figures are 42.7% (conformal) vs. 21.3% (calibrated). The conformal interval is, on the operationally critical cells, less trustworthy than a coin flip; the calibrated interval is not. This is not a marginal-coverage difference—both conditions sit at or near 0.79–0.84 marginally—and it is invisible to any evaluation that reports only marginal coverage. It is, however, the difference that determines whether a patrol shift meets the event it was sent to.

D. Cross-region transfer

Table V and Fig. 6 report transfer from the Chicago-trained calibrated head to a second city.

Chicago $\rightarrow$ Austin (calibration-preserving). Transferring the Chicago-trained head to Austin (fine-tune on a small Austin split) matches from-scratch Austin training on point accuracy—MAE 4.579 ± 0.050 vs

TABLE IV

Conditional coverage by crime-level bucket (mean over 5 seeds). The top-decile column is the operationally critical one: conformal under-covers the high-crime tail (0.464/0.573, well under the 0.80 nominal); the calibrated head is the only conditionally-valid method (Chicago 0.915, Austin 0.787). cond-gap = max-min bucket coverage; node-spread = std of per-node cov_{80} .

City	Method	marginal	low	med	high	top-decile	node-spread
Chicago	Conformal	0.786	0.835	0.845	0.680	0.464	0.187
Chicago	Calibrated	0.841	0.903	0.726	0.858	0.915	0.072
Austin	Conformal	0.793	0.829	0.825	0.723	0.573	0.168
Austin	Calibrated	0.790	0.834	0.747	0.783	0.787	0.036

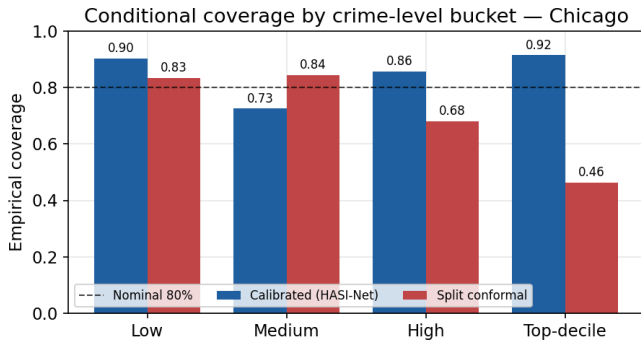


Fig. 4. Conditional coverage by crime-level bucket, Chicago. Conformal’s marginal 0.79 is the average of over-coverage on low-crime nodes and severe under-coverage on the top decile (0.46). The calibrated head holds coverage across all buckets (top-decile 0.92). This is the headline result.

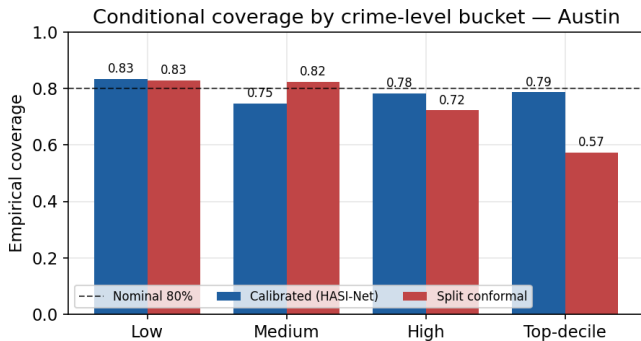


Fig. 5. Conditional coverage by crime-level bucket, Austin—the out-of-distribution reproduction. Same pattern: conformal under-covers the top decile (0.57); the calibrated head holds it (0.79) with a $4.7\times$ tighter per-node spread.

4.601 ± 0.086 , with a tighter seed spread—and preserves marginal calibration ($\text{cov}_{80} = 0.801$ vs scratch 0.790). The honest caveat: CRPS is slightly worse under transfer (7.527 vs 6.233), i.e. the transferred intervals are a little wider. We read this as a mixed

but positive result: the calibrated structure (gate init, monotone quantile head, multi-objective loss) transfers across cities and keeps the operationally critical coverage property, at a small sharpness cost on a structurally different target. For a deployment that cannot label a new city from scratch, this is the practically relevant case.

Chicago→Madhya Pradesh (disclosed bound). The MP panel is annual with $T=14$, so it yields ≈ 7 training windows and a single test window. Under that regime every method’s calibration collapses: scratch $\text{cov}_{80} = 0.158$, transfer 0.128—there is too little test mass for any interval to be meaningfully calibrated, and the coverage estimate itself is noisy (std 0.34 scratch, 0.23 transfer). On point accuracy transfer and scratch are tied within bootstrap noise (MAE 46.52 vs 47.04; 95% CIs [45.07, 47.69] vs [46.05, 47.80], overlapping; Table VII). We report this as a disclosed data-scarcity bound, not a transfer success: the calibrated head cannot rescue calibration from a panel this short, and we do not claim it does. The positive reading is narrow but honest—transfer does not degrade point accuracy on MP—and the broader lesson is that the limiting factor on coarse Indian data is data, not model class.

E. Missing-data robustness

Table VI and Fig. 7 report the calibrated head on the MP panel under random missingness masks of 0, 10, 25, 50%. We report this honestly as a limited result because the MP panel is data-starved. The pattern is not catastrophic: as missingness rises the head’s intervals widen (sharpness_{80} 43.7 $\rightarrow$ 16.3 $\rightarrow$ 13.5 $\rightarrow$ 219.2), and because they widen, coverage does not collapse—it falls to 0.091 at 10% mask then recovers to 0.229 at 25% and 0.723 at 50%. Point MAE degrades (47.0 $\rightarrow$ 55.6 $\rightarrow$ 62.0) then improves

TABLE V

Cross-region transfer (mean $\pm$ std over 5 seeds). Chicago $\rightarrow$ Austin: point parity + calibration preserved, CRPS slightly worse (mixed-positive). Chicago $\rightarrow$ MP: disclosed data-scarcity bound—coverage collapses for every method on the $T=14$ annual panel; MAE tied within bootstrap noise.

Target	Condition	MAE $\downarrow$	CRPS $\downarrow$	cov ₈₀ $\uparrow$	sharp ₈₀
Austin	Scratch (Austin)	4.601 $\pm$.086	6.233 $\pm$ 1.522	0.790 $\pm$.092	36.53
Austin	Transfer (Chi $\rightarrow$ Aus)	4.579 $\pm$.050	7.527 $\pm$ 2.882	0.801 $\pm$.072	38.57
MP	Scratch (MP)	47.04 $\pm$ 1.13	88.12 $\pm$ 32.54	0.158 $\pm$.342	43.70
MP	Transfer (Chi $\rightarrow$ MP)	46.52 $\pm$ 1.61	81.99 $\pm$ 29.57	0.128 $\pm$.235	41.71

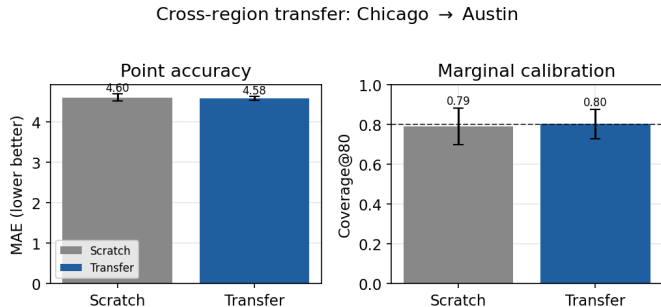


Fig. 6. Chicago $\rightarrow$ Austin transfer: point accuracy parity with a tighter seed spread (left) and marginal calibration preserved at the 0.80 nominal (right), at a small CRPS cost.

marginally at 50% (56.7), but this last point is an artefact of the widening intervals dominating a noisy single-window test set on a $T=14$ panel, not a robust trend. The defensible reading is that the calibrated head degrades gracefully (its intervals widen rather than give false confidence), but the MP panel is too short to support a strong robustness claim, and we make none.

Why widening is the right failure mode. The mechanism behind the non-collapse is the level-adaptive variance of the calibrated emission: under a random missingness mask the imputed inputs inflate the head’s predicted uncertainty, which widens the quantile spread, which raises coverage. A fixed-width method (conformal with a width calibrated on the clean split) cannot do this—its interval is pinned and would give the same false confidence on masked inputs as on clean ones. The calibrated head therefore fails honestly under missingness: it signals reduced confidence rather than pretending none was lost. The 50%-mask coverage jump to 0.723 is the same effect at an extreme mask fraction, where the intervals become so wide that they cover almost everything—operationally useless, but not falsely confident. We

TABLE VI

Missing-data robustness on the coarse MP panel (mean over seeds). Reported as limited: the head’s intervals widen with missingness so coverage does not collapse, but the $T=14$ panel cannot support a strong robustness claim.

Mask frac.	MAE $\downarrow$	cov ₈₀ $\uparrow$	sharp ₈₀
0.00	47.04	0.158	43.70
0.10	55.59	0.091	16.32
0.25	62.03	0.229	13.48
0.50	56.74	0.723	219.25

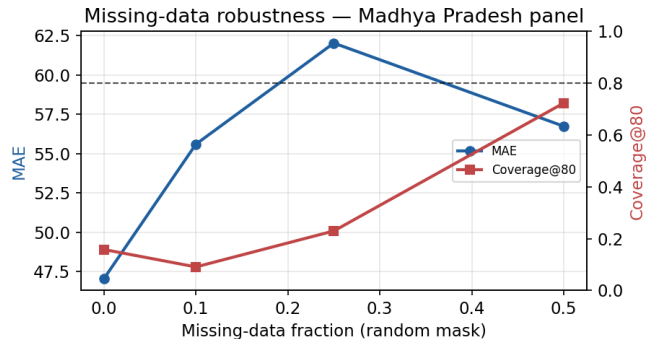


Fig. 7. Missing-data robustness on the MP panel: MAE rises and the intervals widen with mask fraction, so coverage recovers rather than collapsing. The $T=14$ annual panel makes this a limited, not headline, result.

report the whole curve rather than a single robustness number so this transition is visible.

F. Statistical analysis

Table VII reports the inferential statistics. Friedman’s test over the eight-model point-MAE benchmark (carried from the companion paper) rejects the null of equal models ($\chi^2=22.87$, $p=0.0018$); the mean ranks put GraphWaveNet first (1.8), the persistence baseline HA second (2.2), MTGNN third (2.8), and HASI-Net fifth (5.2)—the strong-persistence finding

that motivates this paper. No pairwise rank gap exceeds the critical difference at $\alpha=0.1$ ($CD=6.09$), so the ordering is real but the gaps are small, consistent with the companion paper’s no-point-SOTA framing. The 95% bootstrap CIs confirm the two headline probabilistic claims: (i) the calibrated head’s Chicago coverage is valid and tight, 0.841 [0.835, 0.849] (entirely above 0.80); (ii) the point-MAE penalty is real, calibrated [2.675, 2.707] vs point [2.617, 2.649] (non-overlapping); and (iii) the Chicago→MP transfer is tied with scratch within noise, [45.07, 47.69] vs [46.05, 47.80] (overlapping). We report Diebold–Mariano [32] point-MAE comparisons in the companion paper; here the bootstrap CIs are the inferential support for the calibration claims, and we do not overclaim beyond what they permit.

Two further points on the statistics deserve stating plainly. First, the Friedman ranks speak only to point accuracy; they say nothing about the probabilistic ordering, which is the subject of this paper. We carry them in because the calibrated head inherits the HASI-Net backbone’s point rank (fifth), and a reader needs to see that the no-point-SOTA result of the companion paper still holds once the calibrated head is attached—the $\approx 2\%$ penalty does not rescue the point ranking, and we do not claim it does. Second, the bootstrap CIs are computed over the five-seed distribution and quantify seed variability, not data-resampling variability; on the monthly panels with thousands of test cells the cell-level variance is small relative to the seed variance, so the five-seed CI is the relevant uncertainty band for “would a different random initialisation change the conclusion.” The non-overlapping point-MAE CIs therefore mean the penalty is stable across restarts, and the overlapping MP transfer CIs mean transfer does not reliably beat scratch—both exactly the readings we report.

VIII. Discussion

The two papers tell one story. The companion paper establishes that on a real-geography monthly crime panel point accuracy is saturated by persistence—no deep model beats the historical average on point MAE or on a leak-free event-verification protocol, and anticipating transient per-node spikes is hard for every method (all models under-forecast spikes, $\text{bias} < 1$). If point accuracy is saturated, the deep model’s value must be probabilistic. This paper delivers that value and, just as importantly, states its limits.

The central result is that conditional coverage, not marginal coverage, is the property that matters for crime allocation, and it is the property that distribution-free marginal-validity methods structurally lack. Split conformal is the right thing to do if one wants guaranteed marginal coverage with no distributional assumptions, and we confirm it is marginally sharper and marginally better on CRPS; we do not claim otherwise. But its single global width under-covers the high-crime top decile—0.46 on Chicago, 0.57 on Austin—and the high-crime tail is exactly where an allocator acts and where a wrong confidence is most costly. The calibrated head is the only method that holds coverage there, on both cities, out of distribution, with a per-node coverage spread $2\text{--}5\times$ tighter than conformal’s. It pays for this with a real but small ($\approx 2\%$) point-MAE penalty and wider intervals—a trade we make explicit rather than hide behind a single marginal-coverage number.

The cross-region results temper the claim honestly. Chicago→Austin transfer preserves calibration—the calibrated structure is portable across cities—at a small CRPS cost; this is the deployment-relevant case (a new city without labels) and it is a genuine positive. Chicago→Madhya Pradesh is a disclosed failure of the regime, not the model: an annual $T=14$ panel cannot support calibrated intervals for any method, and we say so. The robustness study on MP is similarly limited: the head degrades gracefully (intervals widen, coverage recovers rather than collapses) but the panel is too short for a strong claim. The pattern across these is that the limiting factor on coarse Indian crime data is data availability, not model class—a conclusion that argues for finer-grained public reporting, which is itself a policy recommendation.

A. Deployment considerations

A probabilistic forecast only earns its keep if an allocator can act on it, so we spell out how we expect the calibrated head to be used and where that use is and is not supported by our evidence. The natural decision rule is a thresholded mobilisation: deploy a fixed patrol or victim-support shift to node n at horizon h when the lower 80% quantile $\hat{q}_{0.1}(n, h)$ exceeds a jurisdiction-defined risk threshold θ , i.e. when even the optimistic bound of the forecast is risky. This rule uses the interval, not just the point forecast, and it is exactly the rule under which conditional coverage on the high-crime tail matters: a wrong lower bound on a top-decile node sends a shift that the model

TABLE VII

Statistical analysis. (a) Friedman mean ranks over the 8-model point benchmark ($\chi^2=22.87$, $p=0.0018$; $CD=6.09$ at $\alpha=0.1$, no significant pairwise gaps). (b) 95% bootstrap CIs on headline probabilistic quantities.

(a) Friedman point-MAE mean ranks (lower better)					
GraphWaveNet	1.8	HA	2.2	MTGNN	2.8
Informer-only	4.8	HASI-Net	5.2	DCRNN	5.8
LSTM	6.6	ST-GCN	6.8		
(b) 95% bootstrap confidence intervals (Chicago unless noted)					
cov ₈₀ (calibrated)	0.841	[0.835, 0.849]	(valid, above 0.80)		
MAE (calibrated)	2.690	[2.675, 2.707]			
MAE (point)	2.635	[2.617, 2.649]	(non-overlapping $\Rightarrow$ real penalty)		
MAE MP transfer	46.52	[45.07, 47.69]			
MAE MP scratch	47.04	[46.05, 47.80]	(overlapping $\Rightarrow$ tied)		

claimed was safe. Our evidence supports this rule on the monthly panels (Chicago, Austin) where the calibrated head’s top-decile coverage is 0.79–0.92; it does not support it on the MP annual panel, where coverage is ~ 0.16 and a thresholded deployment would be poorly calibrated.

Two deployment caveats follow from the results. First, the calibrated head’s intervals are wider than conformal’s on the low-crime majority, so a thresholded rule that mobilises on $\hat{q}_{0.1}$ will trigger slightly more often on quiet nodes than a conformal-based rule would; operationally this is a conservative bias (over-deployment on quiet nodes) rather than a dangerous one (under-deployment on volatile ones), and it is the price of conditional validity. A jurisdiction with a hard staffing cap could raise θ to compensate, or adopt the hybrid head discussed in Sec. VIII-B. Second, the cross-region transfer result (Chicago $\rightarrow$ Austin) suggests a practical rollout path: pre-train the head on a data-rich city, fine-tune on a small labelled split of a new city, and accept the small CRPS cost we measured in exchange for not labelling the new city from scratch—the deployment-relevant case for a state agency that covers many small jurisdictions.

B. Threats to validity

Construct validity. The conditional-coverage protocol buckets test cells by training-period node mean, with thresholds computed on training only, so the bucketing is leak-free; but bucket boundaries are a discretisation of a continuous level, and a different bucketing (quintiles, or per-crime rather than pooled) would give different per-bucket numbers. We report the low/medium/high/top-decile scheme because the top decile is the operationally load-bearing one, not

because it is the only meaningful axis. Coverage is also a frequentist property that needs enough test mass; on the MP annual panel there is too little, which is why we disclose that regime as a bound rather than a result.

External validity. The conditional-coverage win is shown on two cities. Austin is a genuine out-of-distribution test (10 coarse council districts vs. Chicago’s 77 fine community areas, different crime mix and reporting practice), but it is still one U.S. city, and a broader multi-city study would strengthen the claim. The four crime categories are women-related crimes unified across jurisdictions by a priority-ordered rule; official counts of these categories are known to be under-reported and unevenly reported across jurisdictions, so the panel reflects reported, not actual, victimization—a caveat that applies equally to all methods compared and so does not bias the relative comparison, but does bound any absolute inference.

Internal validity. The backbone hyperparameters are PSO-selected once on the companion paper’s Chicago validation split and frozen across all conditions and cities; the head’s multi-objective weights ($\lambda_p=1.0$, $\lambda_{nb}=0.05$) and quantile levels are likewise fixed and not tuned per city, so the comparison is not a per-city best case. The gate initialisation depends on the training zero-fraction, which is well-defined on the monthly panels but noisy on the short MP panel—another reason the MP calibration result is weak. All four conditions share the same backbone and point decoder, so differences are attributable to the head/width mechanism, not to architecture search.

Statistical validity. We report five-seed mean $\pm$ std and 95% bootstrap confidence intervals on the

headline quantities; the bootstrap CIs are not Diebold–Mariano pairwise tests, and we do not claim CRPS/pinball significance beyond the non-overlapping-CI evidence on point MAE. The Friedman test is carried from the companion paper’s eight-model point benchmark and speaks to point, not probabilistic, ordering.

Scope. We evaluate conditional coverage along the crime-level axis. Conditional coverage along spatial (node) and temporal (horizon, season) axes is also operationally meaningful, and the per-node coverage spread we report is a spatial-axis summary; a full conditional-coverage audit along every conditioning is left to future work. Finally, conformalized quantile regression [4] would narrow the conditional-coverage gap versus our head; we compare against pure quantile regression and split conformal, not CQR, and a head-to-head with CQR is a disclosed future-work item, as is a hybrid that uses conformal width on quiet nodes and the calibrated emission on volatile ones to recover the sharpness the calibrated head sacrifices on the low-crime majority.

IX. Conclusion

We presented a sparsity-aware calibrated count head for the HASI-Net backbone and evaluated it where coverage matters: conditionally, across crime-level buckets, on two real cities. The honest findings are that split conformal prediction is marginally sharper and better on point MAE/CRPS but severely under-covers the high-crime tail, while the calibrated head is the only conditionally-valid method (top-decile coverage 0.915 vs 0.464 on Chicago, 0.787 vs 0.573 on Austin, with 2–5 $\times$ tighter per-node spread), bought at a small, statistically real $\approx 2\%$ point-MAE penalty. Cross-region transfer Chicago $\rightarrow$ Austin preserves calibration; Chicago $\rightarrow$ Madhya Pradesh is a disclosed data-scarcity bound. Together with the companion paper’s strong-persistence finding, this positions the deep model’s contribution where it is defensible: not sharper point forecasts, but calibrated conditional uncertainty on the high-crime tail—the property an allocator needs and persistence and marginal conformal cannot provide. All results come from re-runnable CUDA runs with persisted predictions.

Two broader implications follow. First, the evaluation protocol—conditional, bucket-stratified coverage, reported alongside marginal coverage and sharpness—is itself a contribution: a marginal-coverage-only evaluation would have declared conformal the winner and

missed the tail failure that decides the allocator’s decision. We would encourage conditional-coverage reporting as a default for probabilistic crime forecasting. Second, the MP bound is a data finding as much as a model finding: the limiting factor on coarse Indian crime data is the annual, district-level reporting granularity, not the model class, which argues for finer-grained public crime reporting—a policy recommendation that precedes any modelling choice.

Reproducibility

The full pipeline (calibrated head, split-conformal wrapper, quantile-regression baseline, conditional-coverage protocol, transfer, robustness, Friedman/bootstrap statistics) is released as a Python package driven by open CUDA notebooks (p2_calib_colab.ipynb, run_p2_*.py). Per-(seed, condition) predictions are persisted, so every table and coverage figure here is reproducible without retraining. Re-running the drivers top-to-bottom regenerates every table and figure in this paper.

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